خدمتكم طريق خضناه لرضى الله

2025

حلول تجارب

مختبر الكيمياء العامة



لجنة السنافر

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Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 2

Physical Properties: part 1 (Density and Solubility)

Student Name:

Student Number:

Submission Date:

Data & Calculation:

Solubility:

*The table below for testing solute with solvents:

Solvent \ Solute	NaCl	I_2	Glucose	Hexane (C_6H_{12})
H_2O	Soluble	insoluble	soluble	insoluble
Hexane(C_6H_{12})	insoluble	soluble	insoluble	–

Part A: the density of an irregular shaped solid:

Unknown NO.: _____

Mass of the solid	16.1818
Initial volume of water	10 mL
Final volume of water	12.5 mL
Volume of solid	2.5 mL
Density of solid	6.47 g/cm³

Show Calculation:

$$\rho = \frac{m}{V} = \frac{16.1818}{\frac{5}{2}} = 6.47\ g/cm^3$$

Part B: density of unknown liquid.

Unknown NO.: _____

Mass of empty volumetric flask / Pycnometer	42.46 g
Mass of volumetric flask / Pycnometer + liquid	91.859
Mass the unknown liquid	49.39 g
Volume of unknown liquid	50 mL
Density of liquid	0.9878

Show Calculation:

$m_{liquid} = m_2 - m_1 = 91.85 - 42.46 = 49.39$

$\rho = \frac{m}{V} = \frac{49.39}{50} = 0.9878\ g/mL$

Questions

1) Which of the following compounds is water soluble? Why?
   A) K_2SO_4 soluble
   B) C_2H_5OH soluble
   C) $CH_3(CH_2)_{10}CH_3$ insoluble
   D) $CaCO_3$ insoluble

2) In which of the following solvents KCL is dissolve? Explain your answer.
   A) H_2SO_4 insoluble sol.
   B) HCl soluble
   C) C_6H_6 Benzene insoluble

Explanation:

3) What is the affect if you introduced a large quantity of compound into test tube during the experiment of solubility?

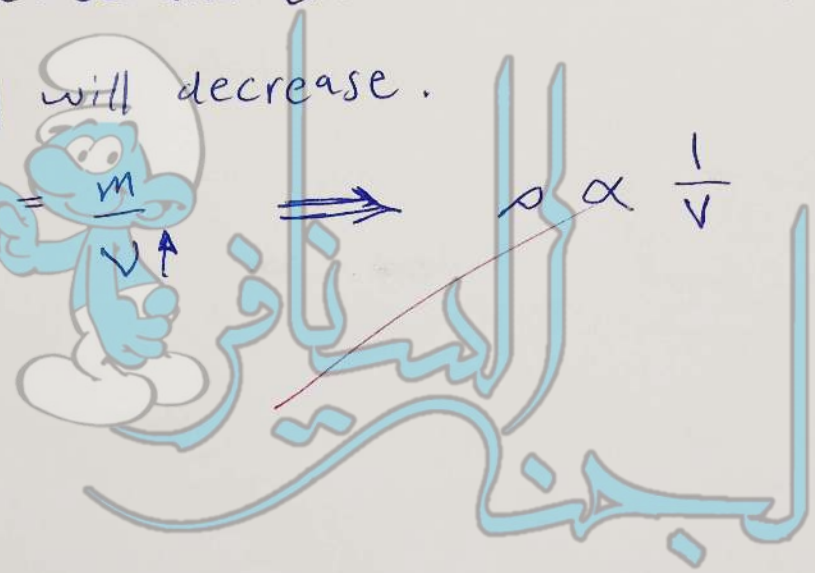
When a large quantity of a compound is put into a test tube, the compound is going to be more saturated & the remains will precipate.

4) Will the measured density of a solid material be affected if a part of the solid dissolved in the liquid used and the volume increased?

Explanation:

Yes, because when the volume increase, the density will decrease.

$\downarrow \rho = \frac{m}{V\uparrow} \Longrightarrow \rho \propto \frac{1}{V}$

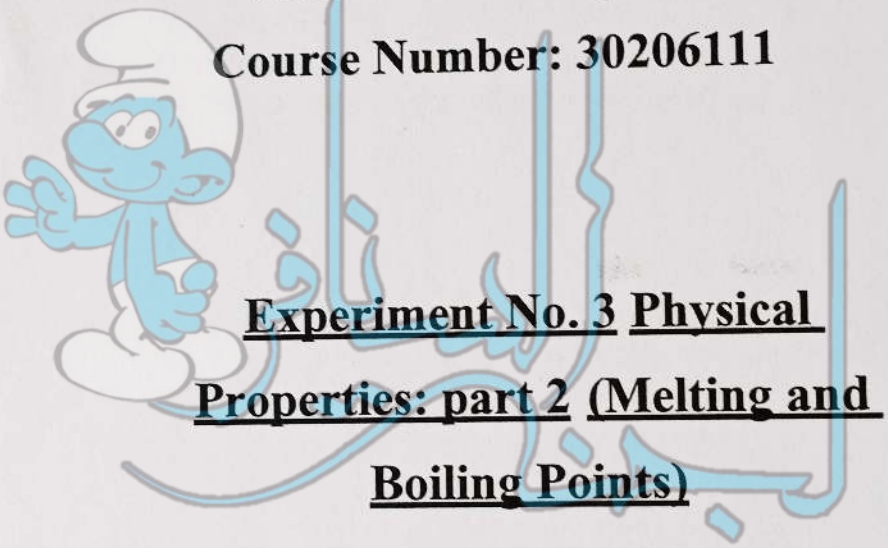


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Experiment No. 3 Physical Properties: part 2 (Melting and Boiling Points)

Student Name:

Student Number:

Submission Date:

Data & Results:

The Melting point of unknown: ----------

The Boiling point of unknown: ~~61 C°~~ ~~76~~ 67 C°

Questions

1) What is the effect of each following factors on the measured melting point of a solid (increase, decrease, has no affect) & give good explanation for each answer?

a- Rapid heating:

b- A large quantity of compound was introduced into the capillary tube:

c- Impurities:

2) What is the effect of each following factors on the measured boiling point of a liquid (increase, decrease, has no affect)?

a- If the test tube base and the thermometer bulb are not at the same level:

→ If the thermometer is higher than the test tube ⇒ decrease

→ If the thermometer is lower than the test tube ⇒ increase

b- If there's some glass shards with liquid sample:

has no effect.

c- If the atmospheric pressure increase:

Increase

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Experiment No. 4

Formula of Hydrated Salts

Student Name:

Student Number:

Submission Date:

Data and calculation:

Unknown No:	Experiment	Duplicate X
(a) Mass of empty crucible m_1	23.4141 g	G
(b) Mass of crucible and hydrate m_2	24.193 g	G
(c) Mass of crucible and anhydrous بعد التسخين والتبريد : m_3	23.831 g	G
(d) Mass of hydrate [(b) – (a)] $m_2 - m_1$	0.7789 g	G
(e) Mass of anhydrous [(c) – (a)] $m_3 - m_1$	0.4176 g	G
(f) Mass of water[(d) – (e)] $m_2 - m_3$	0.3613 g	G
(g) Percentage of water in hydrate [(f) ÷ (d)] × 100	46.38 %	%
(h) Moles of water [(f) ÷ molar mass (H_2O) =18 g/mol]	0.02 mole	Mole
(i) Molar mass of anhydrous (Mm)	258 g/mole	g/mole
(j) Moles of anhydrous [(e) ÷ Mm of anhydrous]	1.6×10^{-3} mole	Mole
(j) Value of X [(h)÷(j)]	12.356	
(k) Formula of hydrate	$KAl(SO_4)_2 \cdot 12H_2O$	

Show Calculations

.....

.....

.....

Questions:

Q1- A hydrated $MgSO_4.XH_2O$ salt, weighing 3.211 g, was heated in a crucible until reaching a constant weight. The weight of the anhydrous $MgSO_4$ was 1.570 grams.

$m_{H_2O} = m_1 - m_2 = 3.211 - 1.57 = 1.641\ g$

a) Calculate the mass percent of water in the hydrated $MgSO_4$ salt.

$$H_2O\% = \frac{m\ H_2O}{m\ Hyd} \times 100\% = \frac{1.641}{3.211} \times 100\% = 51.1\ \%$$

b) Calculate the moles of H_2O removed and the moles of anhydrous $MgSO_4$ remaining in the crucible.

$$\text{moles of } H_2O = \frac{mass}{Mm} = \frac{1.641}{18} = 0.091\ mol$$

$$\text{moles of } MgSO_4 = \frac{mass}{Mm} = \frac{1.57}{120} = 0.013\ mol$$

c) What is the formula for the hydrated $MgSO_4$?

1 mol Hyd : 1 mol anhy : X moles H_2O

1 mol anhy $\longrightarrow$ X H_2O

0.013 mol $\longrightarrow$ 0.091 mol

$$X = \frac{0.091 * X}{0.013} = 7$$

$MgSO_4 7H_2O \longrightarrow MgSO_4 + 7H_2O$

Q2- In today's experiment what error in the data is likely to occur if the hydrated salt is heated too strongly?

~~the anhydrous~~ due to the strong amount of heat ~~the hydrated salt~~ it will explode and it's mass will decrease ~~from~~ which introduces ~~some~~ errors in calculating X.

Q3- How long should the hydrated salt be heated in removing the water, and what happened on the value of X if incomplete dehydration is occur?

It should be heated for 8 mins, the value of X will decrease

$$H_2O\% = \frac{32.8}{100}\% = \frac{m\ H_2O}{m\ Hyd} \times 100\%$$

- $m\ H_2O$ — mole
- $m\ Hyd.$
- $m\ anhy$ — mole

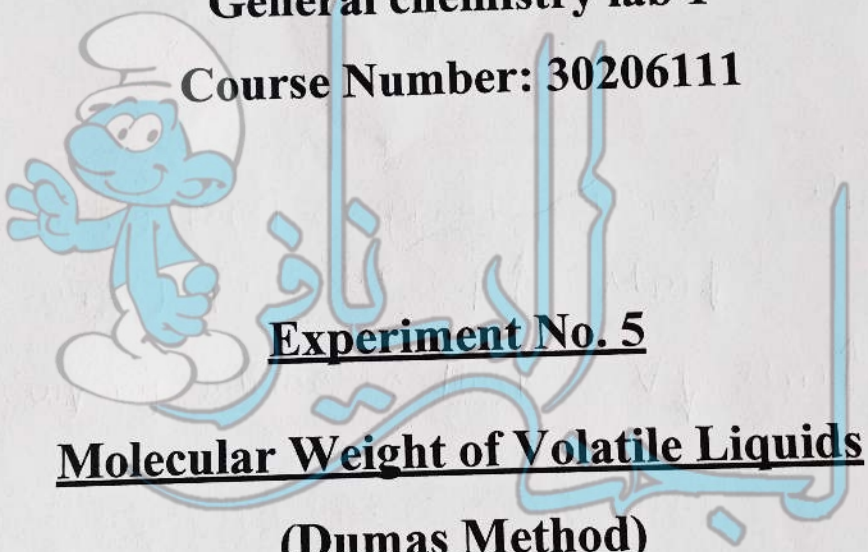
$$X = \frac{m\ H_2O}{m\ anhy}$$

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Experiment No. 5

Molecular Weight of Volatile Liquids

(Dumas Method)

Student Name:

Student Number:

Submission Date:

Data and Calculations:

Mass of empty flask + aluminum foil + rubber band (m_1)	47.2028g
Mass of empty flask + aluminum foil + rubber band + condensed vapor (m_2)	47.4055g
Mass of condensed vapor (m_{gas})	0.2027g
T (°C)	88°C
T (K)	361K
Pressure (mmHg)	756mmHg
Pressure (atm)	0.9917atm
Volume of flask (mL)	69 mL
Volume of flask (L)	0.069 L
Molecular weight of volatile liquid (g/mol)	87.55g/mol

Show calculations

$m_{gas} = m_2 - m_1 = 47.4055 - 47.2028 = 0.2027g$

$P(atm) = \frac{756}{760} = 0.9917\ atm$

1 L $\longleftrightarrow$ 1000 mL $\quad X = \frac{69}{1000} = 0.069\ L$

X $\longleftrightarrow$ 69 mL

Questions:

Q1- A sample of an unknown liquid is vaporized in a flask having a volume of 265 mL at 99.9 °C. The vapor has a mass = 0.725 g, and exerts a pressure = 745 torr. Calculate the gram molecular weight of the unknown liquid.

$$M_m = \frac{mg \cdot R \cdot T}{R \cdot V} = \frac{(0.725)(0.08206)(372.9)}{(0.98)(0.265)} = 85.4\ g/mol$$

$$P_{atm} = \frac{P_{torr}}{760} = \frac{745}{760} = 0.98\ atm$$

Q2- Suggest 3 possible sources of experimental error in this experiment.

Error(1): wrong total evaporate: $m_{gas}\uparrow$ $M_m\uparrow$

Error(2): wrong pressure: $P\uparrow$ $M_m\downarrow$

Error(3): wrong volume in flask: $V\downarrow$ $M_m\uparrow$

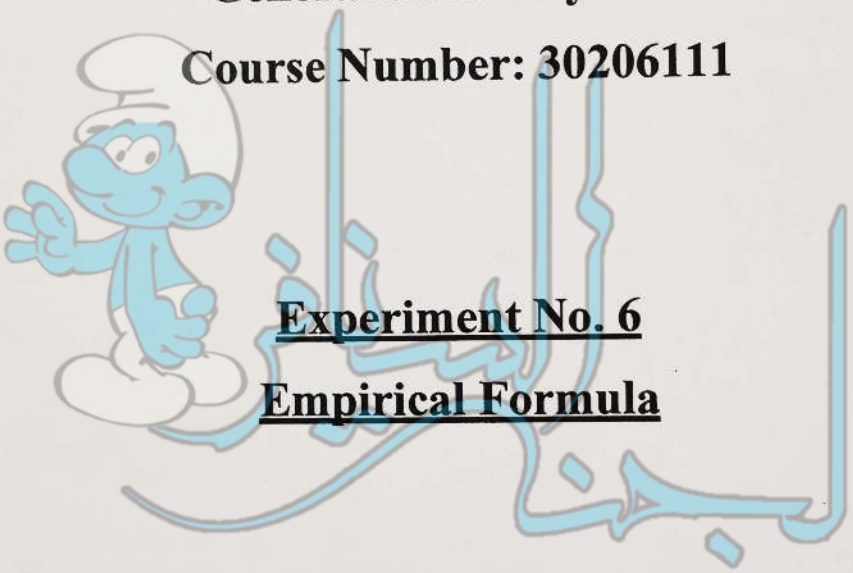
Q3- A student performing this experiment accidently tore the foil cap, so that extra vapor escaped from the flask before it was removed from the heat. Describe how this would affect the mass of vapor and the calculated molar mass.

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Experiment No. 6

Empirical Formula

Student Name:

Student Number:

Submission Date:

Data and Calculation

		Unit
Mass of empty crucible (m_1)	22.1014	g
Mass of crucible + Magnesium (m_2)	22.4905	g
Mass of crucible + Magnesium Oxide (m_3)	22,7218	g
Mass of Magnesium Oxide $m_{Mg_XO_Y}$ (m_3-m_1)	0.6204	g
Mass of Magnesium m_{Mg} (m_2- m_1)	0.3891	~~g~~ ~~mole~~
Moles of Magnesium X =m_{Mg} \24.3 (g\mol)	0.016	mole
Mass of Oxygen m_O =$m_{(Mg X OY)}$ − m_{Mg}	0.2313	g
Moles of Oxygen Y =m_O \16(g\mol)	0.0144	mole
The mole ratio of Magnesium to Oxygen X:Y	1 : 1	~~mole~~
The Empirical formula of Magnesium oxide	MgO	

Show calculation:

1- Moles of magnesium = 0.016 mol

2- Moles of oxygen = 0.0144 mol

3- The Mole ratio = $\frac{0.0144 : 0.016}{0.0144} \rightarrow 1.0 : 1.11$

Questions:

1) How would the calculated molar ratio between (Mg/O) be affect if:

a- Forget the addition of water to get rid of magnesium nitride form.

The Mg/O ratio will ~~decrease~~ increace

b- Incomplete dehydration was done after water was added.

The Mg/O ratio will decreace

c- Some of magnesium oxide spilled out before final weighting.

$m_{MgO} \downarrow \quad m_O \downarrow \quad n_O \downarrow \quad R \uparrow$

2) Calculate Mg% in your compound in this experiment.

$$Mg\% = \frac{0.016}{2.7218} \times 100\%$$

$$\frac{m_{Mg}}{n_{MgO}} = \frac{0.3891}{0.6204} * 100\%$$

$$= 62.7\%$$

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Experiment No. 7

Precipitation Reaction (Limiting Reactant)

Student Name:

Student Number:

Submission Date:

Data and calculations:

** Determination of the limiting reactant:*

Data		**Unit**
Volume of $CaCl_2$	15	mL
Molarity of $CaCl_2$	1.0	M
Volume of Na_2CO_3	20	mL
Molarity of Na_2CO_3	0.7	M
Number of moles of $CaCl_2$ reacted (M x V_L)	0.015	mol
Number of moles of Na_2CO_3 reacted (M x V_L)	0.014	mol
The Limiting reactant is	$CaCl_2$	—
The Excess reactant is	Na_2CO_3	—
The mass of $CaCO_3$ expected (Theoretical yield)	1.401	g
The mass of $CaCO_3$ precipitate (Actual yield) $= m_2 - m_1$	1.828 − 0.414 = 1.414	g
% Yield of $CaCO_3$ $= \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100\%$	100.9%	—

Show calculations

Moles of $CaCl_2$	*Moles of Na_2CO_3*
$n = M * V_L$ $1.0 * 15 \times 10^{-3}$ $= 0.015$ mol	$n = M * V_L$ $= 0.7 * 20 \times 10^{-3}$ $= 0.014$ mol
Moles of $CaCO_3$ expected	*% Yield of $CaCO_3$*
$= n_{Na_2CO_3} = 0.014$ mole	$\frac{1.414}{1.401} \times 100 = 100.9\%$

Questions:

1) If 23 grams of iron (II) chloride ($FeCl_2$) reacts with 41 grams of sodium phosphate (Na_3PO_4), what is the limiting reagent? How much sodium chloride can be formed?

$$3FeCl_2 + 2Na_3PO_4 \longrightarrow Fe_3(PO_4)_2 + 6NaCl$$??

$n_{FeCl_2} = \frac{23}{127} = 0.18\ mol$ | $n_{Na_3PO_4} = \frac{41}{164} = 0.25\ mol$

$\frac{0.18}{3} = 0.06\ mol$ (1 : 1) $\frac{0.25}{2} = 0.125\ mol$

L.R Excess

$3\ mol\ FeCl_2 \longrightarrow 6\ mol\ NaCl$

$0.18 \longrightarrow x$

$x = \frac{0.18 * 6}{3} = 0.36\ mol$

$mass\ NaCl = n_{NaCl} * Mm$

$= 0.36 * 58.5 = 21.06g$

2) How much of the excess reagent remains when this reaction has gone to completion?

Na_3PO_4 (0.25 mol)

React with $FeCl_2$

$3\ FeCl_2 \rightarrow 2\ Na_3PO_4$

$0.18 \rightarrow x$

$x = \frac{0.18 * 2}{3} = 0.12\ mol$

Remain as excess

$0.25 - 0.12 = 0.13$

total moles react

mass remains

$n * Mm$

$= 0.13 * 164$

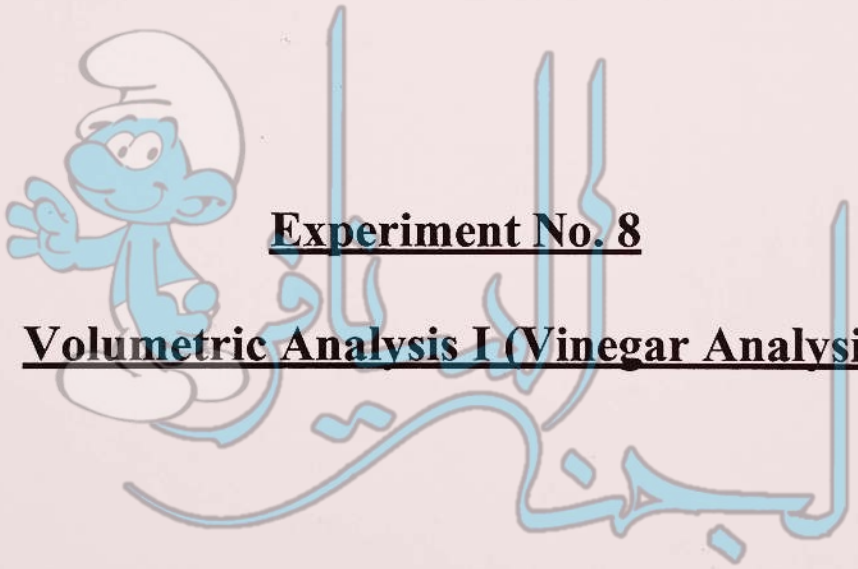
$= 21.32g$

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Experiment No. 8

Volumetric Analysis I (Vinegar Analysis)

Student Name:

Student Number:

Submission Date:

Data and Calculations

		Unit
Volume of diluted vinegar solution	10	mL
Molarity of NaOH solution	0.5	M
Initial burette reading of NaOH solution V_i	9.6	mL
Final burette reading of NaOH solution V_f	19.7	mL
Volume of NaOH solution $(V_f - V_i)$	10.1	mL
Moles of NaOH 0.00505	~~0.0046~~	~~mol~~
Moles of CH_3COOH	0.00505	mol
Mass of CH_3COOH	0.303	g
Molarity of diluted vinegar solution	0.505	M
Mass percentage of CH_3COOH in solution ($\%CH_3COOH$) $= \frac{m_{CH_3COOH}}{m_{vinegar\ sol}} \times 100\%$	2.831 %	—

0.01 L

Show calculations: $M.m_{CH_3COOH} = 60\ g/mol$

$m_{vinger} = 0.505 \times 0.01 = 0.00505$

$n_{vinger} = 0.00505$

$\%CH_3COOH = \frac{0.0107}{0.00505} = \frac{0.303}{10.7} \times 100\%$

$= 2.831\ \%$

$m_{vinger} = V_{vinger} \times$ ~~density~~ $= 10 \times 1.07$

$= 10.79$

QUESTIONS

1. What was the purpose of the phenolphthalein indicator in this experiment?

To check the end point of the experiment.

2. Suppose you added 40 mL of water to your vinegar sample instead of 20 mL. Would the titration have required more, less or the same amount of NaOH solution for a complete reaction? Explain.

ΔV_{NaOH} will not be effected $n_{fixed} = \downarrow M \uparrow V$

therefore n is the same but M changes (less)

3. Suppose you had titrated your vinegar sample with barium hydroxide instead of sodium hydroxide:

ratio: 1 mol ⟷ 2 mol

$Ba(OH)_{2(aq)} + 2CH_3COOH_{(aq)} \rightarrow Ba(CH_3COO)_{2(aq)} + 2H_2O_{(l)}$

Consider a 0.586 M aqueous solution of barium hydroxide, What volume (in mL) of 0.586 M $Ba(OH)_2$ solution are required to neutralize 10 ml of vinegar containing 2.78 g of acetic acid?

$M_{Ba(OH)_2} = 0.586\,M$

$m_{vinger} = 2.78 \longrightarrow n_{vinger} = \frac{2.78}{2.60} = 0.04633\ mol$

$= 0.023165\ mol$

$V_{Ba(OH)_2} = \left(\frac{M_{Ba(OH)_2}}{n_{vinger}}\right)^{-1} = \left(\frac{0.586}{0.02316}\right)^{-1} = \left(\frac{1.172}{0.0463}\right)^{-1}$

$= [25.313]^{-1} = 0.0392\,L$

$= 39.2\ mL.$

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Experiment No. 9

Volumetric Analysis II (Bleach Analysis)

Student Name:

Student Number:

Submission Date:

leads to the number of moles of **iodine** which in turns allows the determination of the moles of **hypochlorite** in the diluted bleach. Finally, the number of **hypochlorite** moles in the original bleach

Data and calculation:

	Trial 1	Trial 2	Unit
Volume of diluted bleach solution	10		mL
Dilution factor	10		
Molarity of $S_2O_3^{-2}$ solution	0.25		M
Initial burette reading of $S_2O_3^{-2}$ solution → brown	26.5		mL
Final burette reading of $S_2O_3^{-2}$ solution → colorless	30.6		mL
Volume of $S_2O_3^{-2}$ solution Vf - Vi (L)	4.1×10^{-3}		L
Moles of $S_2O_3^{-2}$ M * V(L)	1.025×10^{-3}		mol
Moles of I_2 $\frac{1}{2}$ n $S_2O_3^{-2}$	5.125×10^{-4}		mol
Moles of OCl^- = n I_2	5.125×10^{-4}		mol
Molarity of diluted bleach = $\frac{n OCl^-}{V(L)}$	0.05125		M
Molarity of original concentrated bleach = $\frac{n OCl^-}{V org (1mL)}$	0.5125		M

* Mass percentage of NaOCl in original bleach solution (%NaOCl) (show your calculations)

$$\%NaCl = \frac{m\ NaCl}{m\ org\ bleach} \times 100\%$$

$$= \frac{0.038}{1.02} \times 100\% = 3.72\%$$

→ $m\ bleach = \frac{V org}{mL} * \frac{d}{g/mL}$

→ $m\ NaCl = n OCl^- * Mm$

**Mass percentage of Cl_2 in original bleach solution (%Cl_2) (show your calculations)

$$\%\ Cl = \frac{m Cl_2}{m\ bleach} \times 100\% = \frac{0.018}{1.02} \times 100\%$$

$$= 1.78\ \%$$

$$n Cl_2 = \frac{1}{2} OCl^- = \frac{1}{4} n S_2O_3^{-2}$$

$$= 2.5625 \times 10^{-4}\ mol$$

$$m Cl_2 = \frac{1}{2} n OCl^- * M m Cl_2$$

$$= 0.0189$$

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Experiment No. 10

Thermochemistry and Calorimetry

Student Name:

Student Number:

Submission Date:

Example 2: The enthalpy of formation of magnesium oxide is the enthalpy of the following reaction:

$$Mg_{(s)} + \tfrac{1}{2}O_{2\,(g)} \longrightarrow MgO_{(s)} \qquad \Delta H_{form}$$

Directly measuring the enthalpy of this reaction is challenging, however we can determine this enthalpy from the enthalpy of solution of magnesium metal

Data &calculation

Part 1:

	Value	unit
Mass of empty cup-calorimeter m_1	8.669	g
Mass of empty cup-calorimeter + $MgCl_2$ solution m_2	33.65	g
Mass of $MgCl_2$ solution (m_2-m_1)	24.98	g
T_i	24	°C
T_f	30	°C
ΔT	6	°C
Heat gained by the calorimeter $q_{cal.}$	0	
Heat gained by the solution $q_{sol.}$	470.66	J
Heat lost by the reaction $q_{rxn.}$	-470.66	J
Mass of Mg m_{Mg}	0.107	g
Number of moles of Mg n_{Mg}	4.45×10^{-3}	mole
ΔH_1	105.569	J

Part 2:

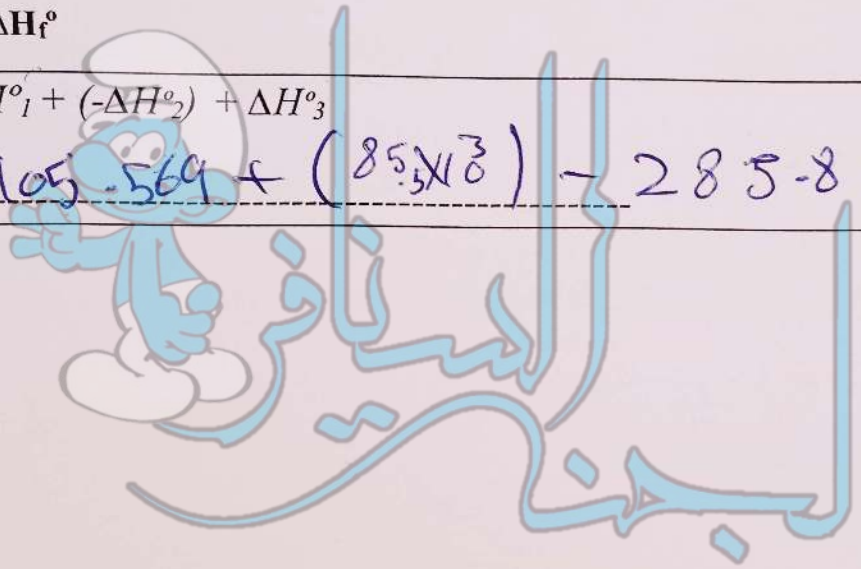
	Value	unit
Mass of empty cup-calorimeter (m_1)	8.7234	g
Mass of empty cup-calorimeter + $MgCl_2$ (m_2)	34.1359	g
Mass of $MgCl_2$ solution (m_2-m_1)	25.4125	g
T_i	26	C°
T_f	30	C°
ΔT	4	C°

Heat gained by the calorimeter	q_{cal}	0	
Heat gained by the solution	q_{sol}	470.6	J
Heat lost by the reaction	q_{rxn}	-470.6	J
Mass of MgO		0.220	g
Number of moles of MgO	n_{MgO}	5.5×10^{-3}	mol
ΔH^{o}_{2}		-85.8×10^{3}	J

Calculation of ΔH_f^o

$\Delta H^{o}_{form} = \Delta H^{o}_{1} + (-\Delta H^{o}_{2}) + \Delta H^{o}_{3}$

$\Delta H^{o}_{form.} = -105.569 + (85.3 \times 10^{3}) - 285.8 = -305.869$



Post lab Questions:

Q1- Use the thermochemical equations given below to calculate the enthalpy change for the following reaction.

$N_2O_{4(g)}$ ------> $2NO_{2(g)}$ $\Delta H = ?$

(1) $N_{2(g)} + 2O_{2(g)}$ ------> $2NO_{2(g)}$ $\Delta H = +66.36$ kJ\mol

(2) $N_{2(g)} + 2O_{2(g)}$ ------> $N_2O_{4(g)}$ $\Delta H = +9.16$ kJ\mol

$N_2O_{4(g)} \longrightarrow N_2 + 2O_{2(g)}$ $\Delta H = -9.16$ KJ/mol

$2O_{2(g)} + N_{2(g)} \longrightarrow 2NO_2$ $\Delta H = +66.36$ KJ/mol

$N_2O_4 \longrightarrow 2NO_2$ $\Delta H = 57.2$ J/mol

Q2- A 113.25 gram sample of gold is initially at 100.0 °C. It gains 20.00 J of heat from its surroundings. What is its final temperature? (specific heat of gold = 0.129 $J\ g^{-1}\ °C^{-1}$)

$q = m * sp * \Delta T$

$\Delta T = \frac{q}{m * sp}$

$= \frac{-20}{113.25 * 0.129} = 1.368$ C°

$\Delta T = T_f - T_i$

=